FRK 2.3.2

Fixed Rank Kriging



The package `FRK` is available on CRAN! To install, please type

```
install.packages("FRK")
```

To install the most recent development version, please load `devtools` and type

```
install_github("andrewzm/FRK", dependencies = TRUE, build_vignette
```

A paper introducing the package is available here. A paper detailing the approach in a non-Gaussian setting is available here (a six-page summary of this paper is available here).

The vignette "FRK_intro" summarises the package, gives details on the EM algorithm that may be employed in a Gaussian setting, and provides several examples. Another vignette, "FRK_non-Gaussian", summarises inference in a non-Gaussian setting (where a Laplace approximation is used), and contains examples using non-Gaussian data and the newly available plotting methods. To access the vignettes, please click on the following links:

Introduction to FRK

Tutorial on modelling spatial and spatio-temporal non-Gaussian data with FRK

A `pkgdown` page is also available here.

Supporting and citing

If you use `FRK` in your work, research, or other activities, please cite it using the information provided by `citation("FRK")`.

FRK 2.3.2

Package: FRK

Type: Package

Title: Fixed Rank Kriging

Version: 2.3.1

Date: 2024-07-16

Author: Andrew Zammit-Mangion, Matthew Sainsbury-Dale

Maintainer: Andrew Zammit-Mangion andrewzm@gmail.com

Description: A tool for spatial/spatio-temporal modelling and prediction with large datasets. The approach models the field, and hence the covariance function, using a set of basis functions. This fixed-rank basis-function representation facilitates the modelling of big data, and the method naturally allows for non-stationary, anisotropic covariance functions. Discretisation of the spatial domain into so-called basic areal units (BAUs) facilitates the use of observations with varying support (i.e., both point-referenced and areal supports, potentially simultaneously), and prediction over arbitrary user-specified regions. `FRK` also supports inference over various manifolds, including the 2D plane and 3D sphere, and it provides helper functions to model, fit, predict, and plot with relative ease. Version 2.0.0 and above also supports the modelling of non-Gaussian data (e.g., Poisson, binomial, negative-binomial, gamma, and inverse-Gaussian) by employing a generalised linear mixed model (GLMM) framework. Zammit-Mangion and Cressie (2021) describe `FRK` in a Gaussian setting, and detail its use of basis functions and BAUs, while Sainsbury-Dale et al. (2024) describe `FRK` in a non-Gaussian setting; two vignettes are available that summarise these papers and provide additional examples.

- Zammit-Mangion, A. & Cressie N. (2021). "FRK: an R package for spatial and spatio-temporal prediction with large datasets." *Journal of Statistical Software*, 98, 1-48. doi:10.18637/jss.v098.i04.
- Sainsbury-Dale, M., Zammit-Mangion, A. & Cressie, N. (2024). "Modelling big, heterogeneous, non-Gaussian spatial and spatio-temporal data using FRK." *Journal of Statistical Software*, 108(10), 1-39. doi:10.18637/jss.v108.i10.

FRK 2.3.2

Quick start

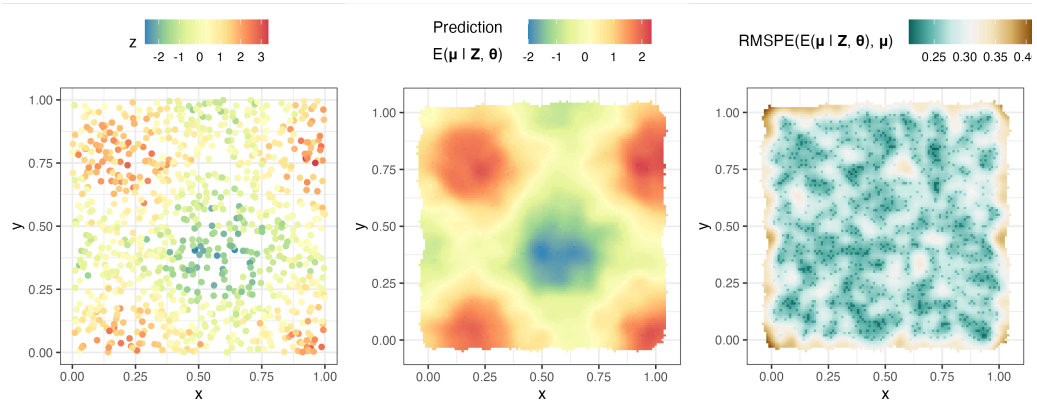
Gaussian data

```
library("FRK")
library("sp")
library("ggplot2")
library("ggpubr")

## Setup
m <- 1000                                                    # Samp
RNGversion("3.6.0"); set.seed(1)                             # Fix
zdf <- data.frame(x = runif(m), y= runif(m))                 # Gene
zdf$z <- sin(8 * zdf$x) + cos(8 * zdf$y) + 0.5 * rnorm(m)    # Simu
coordinates(zdf) = ~x+y                                      # Turn

## Run FRK
S <- FRK(f = z ~ 1,                                  # Formula to FRK
         list(zdf),                                  # All datasets are su
         n_EM = 10)                                  # Max number of EM it
pred <- predict(S)                                   # Prediction stage

## Plotting
plotlist <- plot(S, pred)
ggarrange(plotlist = plotlist, nrow = 1, legend = "top")
```



(Left) Gaussian data. (Centre) Predictions. (Right) Standard errors.

Non-Gaussian data

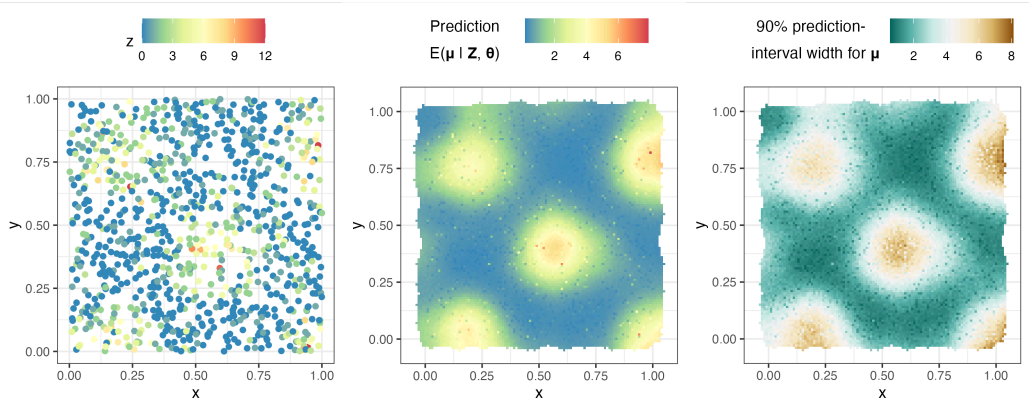
Here we analyse simulated Poisson data. We signify a Poisson data model with a mean response that is modelled using the square-root link

nomial, negative-binomial, gamma, and inverse-Gaussian distributions.

```
## Simulate Poisson data using the previous example's data to cons
zdf$z <- rpois(m, lambda = zdf$z^2)

## Run FRK
S <- FRK(f = z ~ 1, list(zdf),
         response = "poisson",          # Poisson data model
         link = "sqrt")                 # square-root link functio
pred <- predict(S)

## Plotting
plotlist <- plot(S, pred$newdata)
ggarrange(plotlist$z, plotlist$p_mu, plotlist$interval90_mu,
          nrow = 1, legend = "top")
```



(Left) Poisson data. (Centre) Prediction of the mean response. (Right) Prediction interval width of the mean response.

Spatio-temporal data

We now analyse spatio-temporal data, using the NOAA dataset.

```
## Setup
library("spacetime")

data("NOAA_df_1990")
Tmax <- subset(NOAA_df_1990, month %in% 7 & year == 1993)
Tmax <- within(Tmax, {time = as.Date(paste(year,month,day,sep="-"
STObj <- stConstruct(x = Tmax, space = c("lon","lat"), time = "ti

## BAUs: spatial BAUs are 1x1 pixels, temporal BAUs are 1 day inte
```

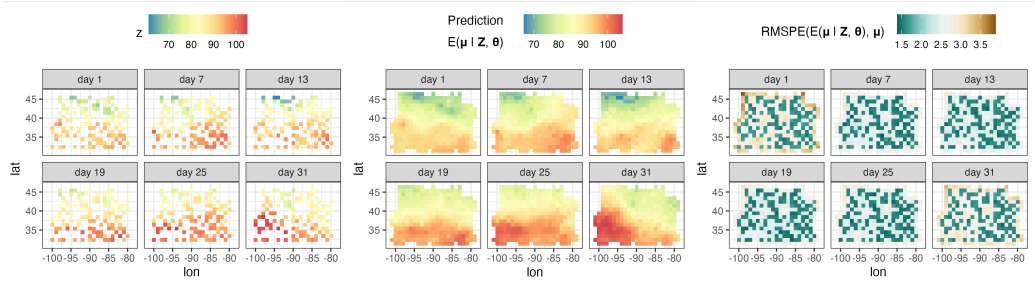
FRK 2.3.2

```
BAUs$fs <- 1 # scalar fine-scale variance matrix, implicit in prev

## Basis functions
G <- auto_basis(manifold = STplane(), data = STObj, nres = 2, tun

## Run FRK
STObj$std <- 2 # fix the measurement error variance
S <- FRK(f = z ~ 1 + lat, data = list(STObj),
         basis = G, BAUs = BAUs, est_error = FALSE, method = "TMB
pred <- predict(S, percentiles = NULL)

## Plotting: include only some times via the argument subset_time
plotlist <- plot(S, pred$newdata, subset_time = c(1, 7, 13, 19, 2
ggarrange(plotlist = plotlist, nrow = 1, legend = "top")
```

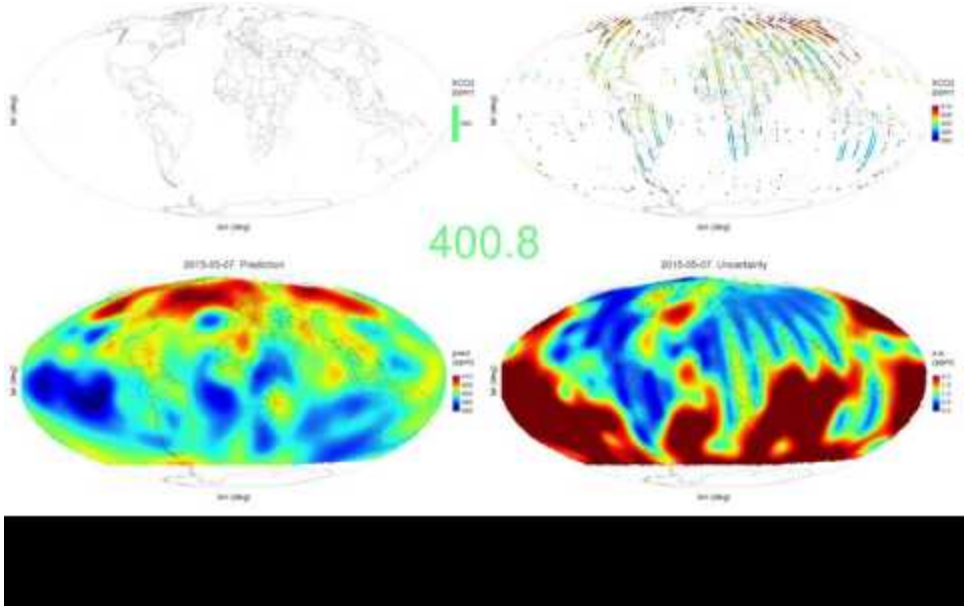


(Left) Prediction of spatio-temporal process. (Right) Prediction interval

Demonstrations

The package `FRK` is currently being used to generate spatio-temporal animations of fields observed by satellite data. Here we show a daily prediction of CO2 using data from the NASA OCO-2 between September 2014 and June 2016.

FRK 2.3.2



Acknowledgements

Thanks to Michael Bertolacci for designing the FRK hex logo!

Links

View on CRAN

Browse source code

Report a bug

License

GPL (>= 2)

Citation

Citing FRK

Developers

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Dev status

CRAN 2.3.2

FRK 2.3.2

Developed by Andrew Zammit-Mangion, Matthew Sainsbury-Dale.

Site built with pkgdown 2.2.0.